Smart-Camera-K15-SL — Control4 Driver

Overview

Complete Control4 driver for Slomins K15-SL IP Camera with full API integration, authentication, streaming, PTZ support, and real-time MQTT event detection.

- **Version:** 1.0.1
- **Package:** `Smart-Camera-K15-SL-v0.1.0.c4z` (22.1 KB)
- **Minimum Control4 OS:** 3.3.2+

Features

Full API Integration

- Initialize camera and retrieve public key
- User authentication with RSA-OAEP encryption
- Token management (temporary and exchange tokens)
- Device listing

Streaming Support

- RTSP main stream (high quality)
- RTSP sub stream (low quality)
- Dynamic URL generation

Camera Functions

- Snapshot URL generation
- PTZ controls (Pan, Tilt, Zoom)
- Preset management

Security

- RSA-OAEP + SHA256 encryption using `C4:Crypto()`
- HMAC-SHA256 signatures for API requests
- Secure token storage

Real-Time Event Detection (MQTT)

- Motion, human, face, and stranger detection
- Doorbell ring notifications with snapshot
- Camera online/offline monitoring

Requirements

- **Control4 Composer Pro** (appropriate version for your platform)
- **Control4 OS** 3.3.2 or newer (for RSA encryption support)
- **Network Access** to API endpoint and camera
- **Camera** accessible on LAN

Installation

1. Install Driver Package

1. Open **Control4 Composer Pro**
2. Go to **Drivers** menu → **Add Driver** → **Install From File**
3. Select `Smart-Camera-K15-SL-v0.1.0.c4z`
4. Click **Install**
5. Driver will appear in available drivers list

2. Add Device to Project

1. Go to **Project** view
2. Select a room
3. Click **Add Device**

4. Search for "Slomins K15-SL"
5. Add to room

3. Configure Properties

API Configuration:

- **Base API URL:** `https://api.arpha-tech.com`
- **Account:** Your email or phone number

Camera Configuration:

- **IP Address:** Camera IP (e.g., 192.168.1.100)
- **HTTP Port:** 80 (default)
- **RTSP Port:** 554 (default)
- **Username:** Camera username
- **Password:** Camera password

Quick Start Guide

```
1. Execute "Initialize"
   → Gets RSA public key from API
   → Stores in "Public Key" property

2. Execute "Login/Register"
   → Encrypts credentials with RSA-OAEP
   → Authenticates with API
   → Stores Auth Token

3. Execute "Get Devices"
   → Lists all devices for user
   → View results in Lua Output

4. Execute "Test Main Stream"
   → Generates RTSP URL for streaming

5. Execute "Get Snapshot URL"
   → Generates HTTP snapshot URL
```

Real-Time Event Detection (MQTT)

The driver supports **MQTT-based real-time event streaming over SSL (port 8884)** and provides notifications for the following events:

- Doorbell ring notifications with snapshot
- Motion detection with snapshot attachment
- Human detection
- Face detection
- Stranger detection
- Camera online/offline status monitoring

MQTT Configuration

Default Behavior

MQTT is **enabled automatically by the driver after authentication**. The driver will automatically:

1. Enable MQTT
2. Fetch MQTT credentials
3. Connect to the MQTT broker
4. Subscribe to camera event topics

You normally **do not need to enable MQTT manually**.

CldBus App Motion Detection Settings

To receive **Motion Detection** and **Human Detection** events, detection must be enabled in the **CldBus mobile app**.

Steps

1. Open the **CldBus App**
2. Select your **camera device**
3. Tap **Settings**
4. Navigate to `Settings → Motion Detection`
5. Under **Detection Type**, select one of the following:

Detection Type	Description
All Detections	All movement events will trigger notifications. This includes general motion detection.
Human Detection	Only human movement will trigger events. Non-human motion will be ignored.

Push Notification Configuration

1. Create Push Notification

1. Open **Composer Pro**
2. Navigate to `Agents → Push Notification`
3. Click **Add Notification** then enter a name for the notification

Configure the following:

Setting	Value
Category	Cameras
Severity	Info / Critical
Subject	Camera Event

Click **Save**.

2. Enable Snapshot Attachment

Edit the created notification and set:

```
Attachment Type = Snapshot URL
```

This allows push notifications to include the **camera snapshot image**.

3. Map Push Notifications in Programming

1. In **Composer Pro**, open **Programming**
2. Select the **Camera Driver** from the device list
3. In the **Events** section, choose the event to trigger (e.g. **Motion Detected**)
4. On the right-side panel, click **Push Notifications**
5. From the dropdown, select the notification you created earlier
6. Drag and drop the notification into the programming area **or double-click the green arrow** to add it

Example:

```
WHEN Motion Detected → THEN Send Push Notification
```

Notification Flow

```
Camera Event (MQTT)
        ↓
Driver Receives Event
        ↓
Driver Processes Event
        ↓
Control4 Event Triggered
        ↓
Push Notification Agent
        ↓
Mobile Notification Sent
        ↓
Snapshot Image Attached
```

Available Actions

All actions accessible via: **Device → Actions → Select action → Execute**

Authentication & API

- **Initialize** — Get RSA public key from API
- **Login/Register** — Authenticate user and get auth token
- **Get Temp Token** — Get temporary token (300 seconds)
- **Get Exchange Token** — Exchange temp token for identity token
- **Get Devices** — List all user devices

Streaming & Snapshots

- **Test Main Stream** — Generate high quality RTSP URL
- **Test Sub Stream** — Generate low quality RTSP URL
- **Test Snapshot** — Generate HTTP snapshot URL

PTZ Controls

- **PTZ Up** — Move camera up
- **PTZ Down** — Move camera down
- **PTZ Left** — Move camera left
- **PTZ Right** — Move camera right

API Functions

1. Initialize Camera

Command: `INITIALIZE_CAMERA`
Endpoint: `POST /api/v3/openapi/init`
Purpose: Retrieve RSA public key for encryption

What it does:

- Generates client ID and request ID
- Creates HMAC-SHA256 signature
- Requests public key from API
- Stores key in "Public Key" property

Expected Output:

```
Camera initialization succeeded
Received public key: MIGfMA...
Public key stored successfully
```

2. Login/Register

Command: `LOGIN_OR_REGISTER`
Endpoint: `POST /api/v3/openapi/auth/login-or-register`
Purpose: Authenticate user with encrypted credentials

Prerequisites: Must run Initialize first (needs public key). Account must be set in properties.

Encryption Process:

```
1. Create crypto object: C4:Crypto("RSA")
2. Import public key
3. Encrypt JSON: {"country_code":"US","account":"user@email.com"}
4. Generate HMAC-SHA256 signature
5. Send to API
```

Expected Output:

```
Using C4:Crypto for RSA-OAEP encryption...
Crypto object created successfully
Public key imported successfully
Encrypting data with RSA-OAEP + SHA256...
Encryption successful!
Login/Register succeeded
Auth token stored
User ID: 12345
```

3. Get Temp Token

Command: `GET_TEMP_TOKEN`
Endpoint: `POST /api/v3/openapi/temperate-token`
Purpose: Get temporary token (300 seconds duration)

Request:

```
{
  "duration": 300
}
```

Response:

```
{
  "code": 20000,
  "message": "success",
  "data": {
    "token": "Em2SB9ijEYrQimb0v8irW/WTOZm67dHHeqArhGBom9M="
  }
}
```

4. Get Exchange Token

Command: `GET_EXCHANGE_TOKEN`
Endpoint: `POST /api/v3/openapi/auth/exchange-identity`
Purpose: Exchange temporary token for identity token

Prerequisites: Must run GET_TEMP_TOKEN first.

Response:

```
{
  "code": 20000,
  "message": "success",
  "data": {
    "data": "Z6jSr30H6P60joVp69nGgTP8pvY4tEodneDAQY6Fe94="
  }
}
```

5. Get Devices

Command: `GET_DEVICES`
Endpoint: `GET /api/v3/openapi/devices-v2`
Purpose: List all devices for authenticated user

Prerequisites: Must run Login/Register first (needs Auth Token).

Headers:

```
Authorization: Bearer <auth_token>
```

6. Test Main Stream

Command: `TEST_MAIN_STREAM`
Format: `rtsp://<ip>:554/streamtype=1`

```
Main Stream RTSP URL: rtsp://192.168.1.100:554/streamtype=1
```

7. Test Sub Stream

Command: `TEST_SUB_STREAM`
Format: `rtsp://<ip>:554/streamtype=0`

```
Sub Stream RTSP URL: rtsp://192.168.1.100:554/streamtype=0
```

8. Get Snapshot URL

Command: `GET_SNAPSHOT_URL`
Format: `http://[user:pass@]<ip>:<port>/snap.jpg`

```
Snapshot URL: http://admin:password@192.168.1.100:80/snap.jpg
```

Properties

Input Properties (Configure These)

Property	Type	Default	Description
Base API URL	STRING	https://api.arpha-tech.com	API endpoint
Account	STRING	—	User email or phone
IP Address	STRING	192.168.1.100	Camera IP
HTTP Port	INTEGER	80	HTTP port
RTSP Port	INTEGER	554	RTSP port
Username	STRING	—	Camera username
Password	STRING	—	Camera password
Snapshot Path	STRING	/snap.jpg	Snapshot path
Enable MQTT	LIST	True	Enable/disable MQTT event streaming

Output Properties (Read-Only/Managed)

Property	Description
Status	Current operation status
Public Key	RSA public key from API
Client ID	Generated client ID
Auth Token	Authentication token
Temp Token	Temporary token
Exchange Token	Identity token
Main Stream URL	High quality RTSP URL
Sub Stream URL	Low quality RTSP URL
Online	Camera online status

Workflow Examples

Workflow 1: Initial Setup & Authentication

```
1. Set "Base API URL" property
2. Set "Account" property (your email)
3. Execute "Initialize"
   → Gets public key
4. Execute "Login/Register"
   → Authenticates user
   → Stores Auth Token
   → MQTT auto-connects after auth
5. Execute "Get Devices"
   → Lists your devices
```

Workflow 2: Token Management

```
1. Execute "Get Temp Token"
   → Gets 300-second token
2. Execute "Get Exchange Token"
   → Exchanges for identity token
3. Use tokens for custom API calls
```

Workflow 3: Streaming Setup

```
1. Set "IP Address" property
2. Execute "Test Main Stream"
   → rtsp://192.168.1.100:554/streamtype=1
3. Execute "Test Sub Stream"
   → rtsp://192.168.1.100:554/streamtype=0
4. Use URLs in Control4 video configuration
   or external VMS
```

Workflow 4: Snapshot Testing

```
1. Set "IP Address", "Username", "Password"
2. Execute "Get Snapshot URL"
```

```
3. URL sent to camera proxy
4. Test URL in browser to verify
```

Workflow 5: Push Notifications

```
1. Complete Workflow 1 (auth + MQTT connected)
2. Create notification in Agents → Push Notification
3. Set Attachment Type = Snapshot URL
4. In Programming, map camera event → Send Push Notification
5. Trigger event to receive mobile notification with snapshot
```

RSA Encryption

The driver uses **Control4's C4:Crypto() API** for RSA-OAEP + SHA256 encryption:

```
-- Create RSA crypto object
local crypto = C4:Crypto("RSA")

-- Import public key (from Initialize)
crypto:ImportPublicKey(publicKey)

-- Encrypt credentials
local encrypted = crypto:Encrypt(data, {
    scheme = "oaep",
    hash = "sha256"
})
```

Requirements

- Control4 OS 3.x or newer
- Public key from Initialize API
- PEM formatted public key

Fallback Mode

If `C4:Crypto()` is unavailable:

1. Set "Pre-Encrypted Post Data" property
2. Driver will use pre-encrypted value
3. For testing only

Troubleshooting

View Logs

1. Open **Composer Pro**
2. Go to **Lua Output** window
3. Execute any action
4. Watch detailed logs: request URLs, headers, bodies, response codes, and error messages

Common Issues

"No public key available"

- Run Initialize first
- Check Base API URL
- Verify network connectivity

"Failed to create C4:Crypto object"

- Control4 OS too old (need 3.x+)
- Use fallback pre-encryption mode

"Login failed"

- Verify Account property is set
- Check public key exists
- Ensure Initialize ran successfully

"IP Address not set"

- Configure IP Address property
- Verify camera is reachable

"No auth token available"

- Run Login/Register first
- Check authentication succeeded

MQTT not receiving events

- Verify "Enable MQTT" is set to True
- Confirm Login/Register completed successfully
- Check motion detection is enabled in CldBus App
- Verify network allows outbound port 8884 (MQTT over SSL)

API Response Codes

Code	Meaning
200 / 20000	Success
401	Unauthorized (invalid token)
400	Bad request (invalid parameters)
500	Server error

Files Structure

```
Smart-Camera-K15-SL/
├── driver.lua              # Main driver logic
├── driver.xml              # Configuration & metadata
├── mqtt_manager.lua        # MQTT event streaming & broker management
├── README.md               # This documentation
├── CldBusApi/              # API helper libraries
│   ├── auth.lua
│   ├── dkjson.lua
│   ├── http.lua
│   ├── sha256.lua
│   ├── transport_c4.lua
```

```
│   └── util.lua
└── build-c4z.ps1           # Build script
```

Building the Driver

Build Package

Run in PowerShell:

```
.\build-c4z.ps1
```

This creates `Smart-Camera-K15-SL-v0.1.0.c4z` ready for installation.

Manual Build

```
$files = @(
    "driver.lua",
    "driver.xml",
    "mqtt_manager.lua",
    "README.md",
    "CldBusApi\*.lua"
)

Compress-Archive -Path $files -DestinationPath "temp.zip"
Rename-Item "temp.zip" "Smart-Camera-K15-SL.c4z"
```

Development Notes

Helper Libraries (CldBusApi/)

- **auth.lua** — Authentication utilities
- **dkjson.lua** — JSON encoding/decoding
- **http.lua** — HTTP request helpers
- **sha256.lua** — HMAC-SHA256 implementation
- **transport_c4.lua** — Control4 transport adapter

- **util.lua** — UUID generation, HMAC functions

Signature Generation

All API requests include HMAC-SHA256 signatures:

```
-- Message format
local message = "client_id=<uuid>&request_id=<uuid>&time=<timestamp>&version=0.0.1"

-- Generate signature
local signature = util.hmac_sha256_hex(message, app_secret)

-- App secret
local app_secret = "hg4IwDpf2tvbVdBGc6nwP5x2XGCIlNv8"
```

UUID Generation

```
local client_id = util.uuid_v4()
-- Example: "0ffb133a-b03d-4cfd-82a9-d45220516199"
```

Version History

v1.0.1 (Current)

- Updated Control4 Camera Programming Options
- **Device Events**: Added "Registered User Detected", "Stranger Detected"
- **Commands Added**: Take Snapshot, Initialize Camera *, Unmute Mic, Mute Mic, Turn up speaker volume, Turn down speaker volume
- **Commands Removed**: Zoom In, Zoom Out, PTZ_MOVE, PTZ_CONT_START, PTZ_CONT_STOP, TEST_SNAPSHOT, TEST_MAIN_STREAM, TEST_SUB_STREAM, BIND_DEVICE, LOGIN_OR_REGISTER, GET_TEMP_TOKEN, GET_EXCHANGE_TOKEN, GET_DEVICES, DISCOVER_CAMERAS, DISCOVER_CAMERAS_SDDP, SET_DEVICE_PROPERTY

- **Conditions Added**: If sensing motion, If not sensing motion, If Mic is muted, If Mic is unmuted, If Speaker volume is 1-10, If tracking is on/off

v0.1.0

- Complete API integration
- RSA-OAEP encryption with C4:Crypto()
- User authentication
- Token management
- Device listing
- RTSP streaming URLs
- Snapshot support
- PTZ controls
- MQTT real-time event detection (SSL, port 8884)
- Push notification support with snapshot attachment

Support

- **Driver Version:** 1.0.1
- **Maintainer:** Slomins
- **Manufacturer:** Slomins
- **Model:** K15-SL
- **Control4 OS:** 3.3.2+
- **API Endpoint:** https://api.arpha-tech.com
- **Website:** www.slomins.com

For issues or questions, check Lua Output logs for detailed error information.

License

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End of Documentation